

Where and How to Fuse: An Ablation Study of IRENE-Style Multimodal Transformers on Public Chest Radiograph, Clinical-Note and Structured Data

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<https://github.com/xiangruiSun/IRENE>

Abstract

IRENE is a multimodal transformer for clinical diagnosis. It treats chest radiographs, unstructured clinical text and structured clinical data as token sequences and fuses them with bidirectional cross-modal attention. Its original evaluation used a private cohort, and the choices of where to fuse, how the modalities attend to each other and whether to add image-text contrastive learning were never isolated. We re-implement IRENE as a single configurable model (IRENE-MM) and study these choices on a fully public multimodal dataset: 782 chest radiographs from 424 patients, each with a free-text clinical note, demographics, vitals and laboratory values, labelled with seven diagnoses. We describe a reproducible data pipeline for image-text-structured triplets that includes text cleaning, masking of diagnosis terms that leak the label, and patient-grouped cross-validation. We then run 29 configurations (330 fold-level trainings). Fusing all three modalities improves on every partial input (macro AUROC 0.794 for late fusion and 0.785 for IRENE, versus 0.757 for note + structured data and 0.693 for images alone); the gain is significant against images alone and borderline against note + structured data. IRENE’s mid-level bidirectional attention matches early fusion, but cross-attention at every layer, or cross-attention without self-attention, is clearly worse (0.741 and 0.752). A CLIP or SigLIP alignment objective does not help diagnosis at this scale. It hurts mid-fusion models (0.726 with CLIP, $\lambda=0.5$) and is neutral for late fusion (0.793), which also gives the best cross-modal retrieval. Every model with fusion loses much of its accuracy when the note is missing at test time. Probing experiments show that the publication source of a case alone reaches 0.740 AUROC, a strong confounder in public COVID-19 collections. We release the code, the per-run logs and this paper.

1 Introduction

Clinical diagnosis is inherently multimodal: a radiologist reads a chest radiograph in the context of the presenting complaint, the history, vital signs and laboratory results. Deep-learning systems that mirror this practice have moved from *non-unified* pipelines, in which each modality is encoded separately and combined only at the decision level, towards *unified* transformers that tokenise every modality and let attention mix them [1–3]. IRENE [4] is a representative clinical instance: chest X-ray patches, chief-complaint tokens, demographics

and laboratory values are all embedded as tokens; the first two layers apply *bidirectional multimodal attention*, in which the image stream and the clinical stream each attend to themselves and to the other stream, and the remaining layers perform joint self-attention over the concatenated sequence.

Three design decisions in such models are rarely isolated experimentally: (i) **where** to fuse (early, mid, or late); (ii) **how** the streams exchange information (bidirectional, unidirectional, co-attention, gated cross-attention); and (iii) whether an explicit **image-text alignment** objective in a shared embedding space (CLIP [5], SigLIP [6]), which dominates 2023–2024 vision-language pre-training [7–10], also helps a supervised clinical fusion model. The original IRENE was trained on a private cohort, so these questions cannot be revisited on its data. This paper does so on a fully public dataset.

Contributions.

- An open, reproducible re-implementation of IRENE for a public multimodal chest-radiograph dataset [11], including an image-text-structured data pipeline (cleaning, de-identification of dates, label-leakage masking, missing-value encoding and patient-grouped splits) and a modernised training stack (native AMP, DDP, tuned DataLoader, W&B tracking) that replaces the original apex/DataParallel code.
- A single configurable architecture, IRENE-MM, that expresses early, mid and late fusion, six cross-modal attention structures and two contrastive objectives as switches, so that ablations differ in exactly one factor.
- A controlled study over 29 configurations with 5-fold patient-grouped cross-validation and multiple seeds, reporting pooled out-of-fold macro AUROC with patient-level bootstrap confidence intervals, missing-modality robustness and cross-modal retrieval.
- Evidence that free-text notes in public COVID-19 collections leak the label, which, if unmasked, inflates text-based and fused models (a multimodal counterpart of the shortcut learning documented for radiographs [12, 13]).

2 Related work

Fusion in clinical multimodal models. Surveys distinguish early (input/feature-level), joint/intermediate and late (decision-level) fusion [1, 2]. MedFuse [14] combines chest X-rays with clinical time series through an LSTM fusion module

robust to missing inputs; IRENE [4] uses a unified transformer with bidirectional multimodal attention followed by joint self-attention. In general vision–language models, single-stream (“merged attention”) and dual-stream co-attention designs such as ViLBERT and LXMERT [15, 16] represent the two ends of the spectrum; attention bottlenecks [17] showed that restricting cross-modal flow to later layers can improve accuracy and cost, and Flamingo [18] introduced tanh-gated cross-attention layers inserted into frozen backbones. Ma et al. [19] showed that multimodal transformers are fragile when a modality is missing at test time and that the optimal fusion strategy is dataset-dependent.

Image–text alignment. CLIP [5] learns a shared embedding space with a symmetric InfoNCE loss [20]; SigLIP [6] replaces the softmax with a pairwise sigmoid loss that does not require global normalisation. ALBEF [21] and BLIP-2 [7] align unimodal representations *before* fusing them with cross-attention. ImageBind [8] binds many modalities into one space through image-paired data. In medicine, ConVIRT [22], GLORIA [23], MedCLIP [24], MGCA [25], MedKLIP [26], CXR-CLIP [10] and BiomedCLIP [9] pre-train radiograph–report encoders contrastively, and generalist models such as LLaVA-Med [27] and Med-PaLM M [28] extend this to instruction following. We study the contrastive objective not as pre-training on millions of pairs but as an auxiliary loss inside a small supervised fusion model, the regime of most hospital-scale datasets.

Shortcuts in COVID-19 imaging. DeGrave et al. [12] and Roberts et al. [13] showed that many COVID-19 radiograph classifiers exploit dataset artefacts. We add that *text* in public case collections carries even more direct shortcuts: diagnosis and pathogen names.

3 Data

Source. We use the COVID-19 Image Data Collection [11] (<https://github.com/ieee8023/covid-chestxray-dataset>), a public set of cases compiled from publications and case repositories. Each image has metadata that naturally splits into IRENE’s three input types: the radiograph, an unstructured `clinical_notes` field (the presenting history and imaging findings, written by many authors in many institutions; several are machine translations of non-English case reports), and structured fields (age, sex, days since symptom onset, temperature, SpO₂, leukocyte, neutrophil and lymphocyte counts, and the radiographic view). Figure 1 summarises the processed data.

Cohort and labels. From 950 rows we keep the 866 radiographs, drop `todo/Unknown` findings, and obtain 782 images from 424 patients. The hierarchical finding string (e.g. `Pneumonia/Viral/COVID-19`) is mapped to 7 targets: COVID-19, other viral, bacterial, fungal, other/unspecified pneumonia (including lipid and aspiration), tuberculosis, and

no finding. Classes are highly imbalanced (Figure 1a); following IRENE the model is trained with multi-label binary cross-entropy and evaluated with per-class and macro AU-ROC.

Text processing. Notes are Unicode-normalised (NFKC), stripped of URLs and figure/panel references, whitespace-normalised, and dates are replaced by a [DATE] token (71 notes contain a date). Crucially, 354 of the 683 notes (52%) contain the diagnosis or pathogen by name (“COVID-19”, “SARS-CoV-2”, “*Pneumocystis*”, “tuberculosis”, ...). Unless stated otherwise we replace 818 such terms by a [DX] token; Section 6.6 quantifies the effect of not doing so. Notes are tokenised with spaCy [29]; each token is represented by a 300-d static word vector plus three indicator channels ([DX], [DATE], out-of-vocabulary), mirroring IRENE’s per-token text features.

Structured data. Continuous fields are robustly standardised (median/IQR, clipped to ± 5) and each value carries an observation mask: laboratory values are missing for over 95% of images (Figure 1c). The view (PA/AP/AP supine/lateral) is one-hot encoded. RT-PCR status and outcomes (survival, ICU, intubation) are excluded because they reveal or post-date the label.

Splits. Many patients contribute several images (up to 8), often with the same note. We therefore use 5-fold *patient-grouped*, label-stratified cross-validation (StratifiedGroupKFold); within each training fold 15% of patients are held out for early stopping. All numbers are computed on out-of-fold (OOF) predictions pooled over the five test folds (782 images), which avoids unstable AUROCs for classes with 3–4 positives per fold.

4 Method

Tokenisation. Figure 2 shows the architecture. Following the “hybrid” tokeniser that IRENE inherits from ViT [30], radiographs are encoded by a DenseNet-121 [31] pretrained on seven public chest X-ray datasets (`torchxrayvision densenet121-res224-all` [32]), kept frozen; its $7 \times 7 \times 1024$ feature map is average-pooled to a 4×4 grid and linearly projected to d -dimensional image tokens with learned position embeddings. Five views per image (one clean, four randomly cropped, rotated and intensity-jittered) are cached; training samples a random view per step. Note tokens (at most 48) are linearly projected; age and sex use IRENE’s scalar embeddings $\text{Linear}(1, d)$, and each structured value v_j with mask m_j becomes $m_j W v_j + (1 - m_j) e_j^{\text{miss}} + p_j$ with a shared projection W , a learned missing embedding and a position embedding.

Dual-stream block. Let $h_I = \text{LN}(\mathbf{X}_I)$, $h_C = \text{LN}(\mathbf{X}_C)$, $S_I = \text{Attn}(h_I, h_I)$, $S_C = \text{Attn}(h_C, h_C)$ and the cross terms $X_{I \leftarrow C} =$

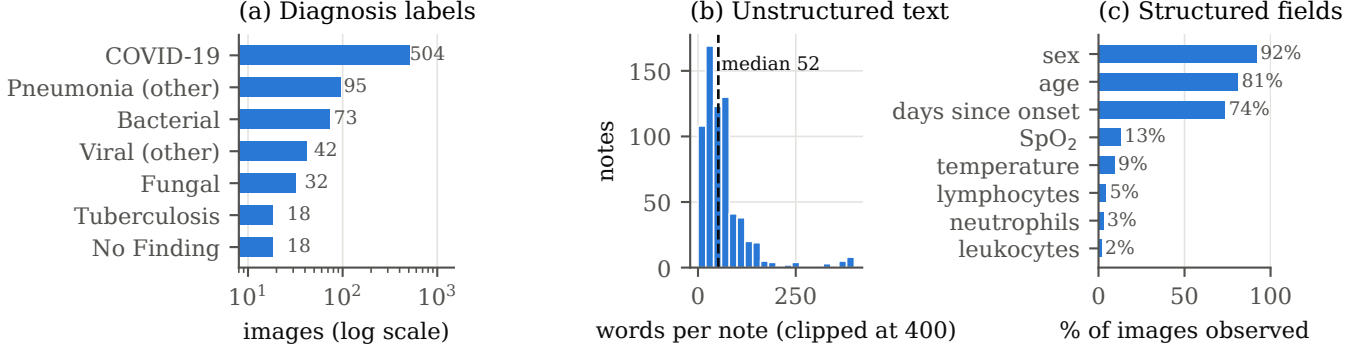


Figure 1: Processed dataset. (a) Label counts (log scale). (b) Length of the unstructured clinical notes (683 of 782 images have one). (c) Fraction of images for which each structured field is observed; missing values are encoded with learned missing-value embeddings rather than imputed.

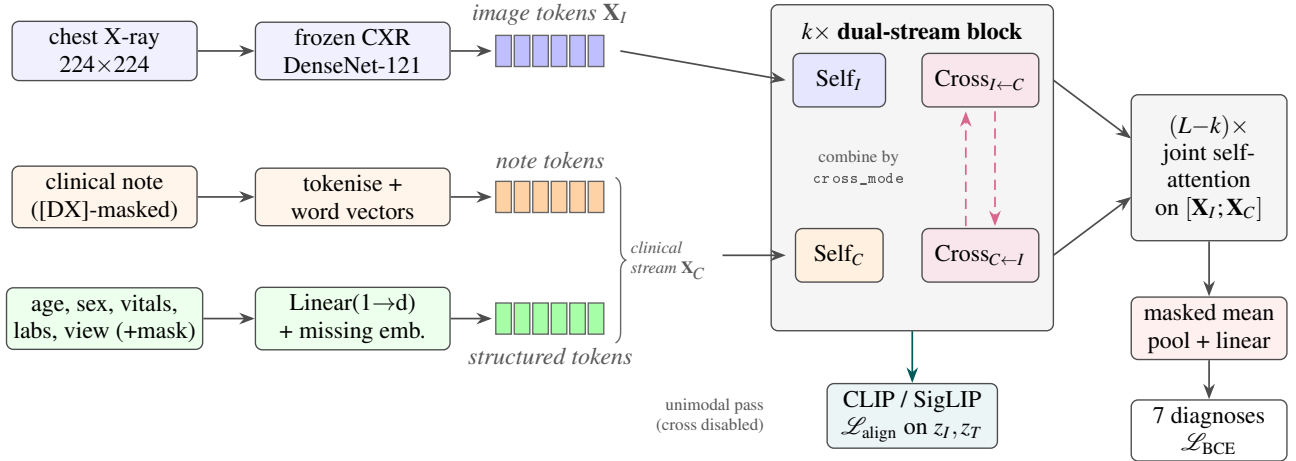


Figure 2: IRENE-MM. Each modality is tokenised; the image stream $\mathbf{X}_I$ and the clinical stream $\mathbf{X}_C$ (note + structured tokens) pass through k dual-stream blocks whose cross-modal exchange is set by `cross_mode`, then through $L - k$ joint self-attention blocks. $k = 0$ is early fusion; $k = L$ with no cross-attention is late fusion; IRENE is $k = 2$ with bidirectional attention. The optional alignment loss is computed on *unimodal* pooled embeddings (a second pass through the dual-stream blocks with cross-attention disabled), so that the note embedding cannot see the image.

$\text{Attn}(q=h_I, kv=h_C)$, $X_{C \leftarrow I} = \text{Attn}(q=h_C, kv=h_I)$, each with its own projections and key-padding masks. The attention outputs A_I, A_C are

cross_mode	A_I	A_C
bi_avg (IRENE)	$(S_I + X_{I \leftarrow C})/2$	$(S_C + X_{C \leftarrow I})/2$
bi_cross (co-attn.)	$X_{I \leftarrow C}$	$X_{C \leftarrow I}$
i2t	$(S_I + X_{I \leftarrow C})/2$	S_C
t2i	S_I	$(S_C + X_{C \leftarrow I})/2$
gated	$S_I + \tanh(\alpha_I)X_{I \leftarrow C}$	$S_C + \tanh(\alpha_C)X_{C \leftarrow I}$
none	S_I	S_C

followed by per-stream output projections, residual connections and feed-forward networks. `bi_avg` is IRENE’s equation (averaging self- and cross-attention contexts); `bi_cross` corresponds to ViLBERT/LXMERT co-attention [15, 16]; `gated` follows Flamingo’s zero-initialised tanh gates [18].

Fusion position. After k dual-stream blocks the streams are concatenated and processed by $L - k$ standard pre-LN blocks [33]. The prediction is a linear head on the masked mean of all tokens (IRENE also mean-pools). With $k = 0$ the model

is a single-stream early-fusion transformer; with $k = L$ and none the streams never interact and their pooled features are concatenated before the head (late fusion).

Contrastive alignment. With projection heads g_I, g_T and pooled unimodal embeddings z_I, z_T (image tokens; note tokens only), the auxiliary loss is either CLIP/InfoNCE [5], $\mathcal{L} = \frac{1}{2} [\text{CE}(\tau S, Y) + \text{CE}(\tau S^\top, Y^\top)]$, or SigLIP [6], $\mathcal{L} = -\frac{1}{B} \sum_{ij} \log \sigma(y_{ij}(\tau S_{ij} + b))$, where $S_{ij} = g_I(z_I^i)^\top g_T(z_T^j)$ on ℓ_2 -normalised vectors and τ (and b) are learned. Because images of the same patient often share one note, Y marks *all* pairs with identical notes as positives (multi-positive targets) instead of treating them as negatives. Following ALBEF [21] we align before fusing; unlike a naive implementation that pools the text stream after cross-attention—where the text embedding already contains the image and retrieval becomes trivial (we observed $R@1 > 0.8$ in a pilot)— z_I, z_T come from a weight-shared pass through the dual-stream blocks with cross-attention switched off. The total loss is $\mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{align}}$.

Missing modalities. All attention uses key-padding masks, so images without a note simply have no note tokens. We evaluate every model a second time with the note removed at test time, and optionally train with *modality dropout* (note removed with probability 0.3).

5 Experimental setup

Model and training. All variants use $L=6$ layers, $d=128$, 4 heads and an MLP width of 256 (1.00–2.39 M trainable parameters depending on the variant; the frozen DenseNet is not counted), dropout 0.1. We train with AdamW [34] ($\text{lr } 3 \times 10^{-4}$, weight decay 0.05, one-cycle schedule), batch size 32, gradient clipping 1.0, at most 25 epochs with early stopping on the validation BCE (patience 5), restoring the best checkpoint. The contrastive projection dimension is 128.

Protocol. Every configuration is run on the same five folds with 3 seeds (main comparison) or 2 seeds (ablations). We report pooled-OOF macro AUROC (mean $\pm$ s.d. over seeds), macro AUPRC, macro F1 and accuracy of the arg-max prediction, a 95% patient-level bootstrap interval of the seed-averaged OOF predictions (1,000 resamples), and paired patient-level bootstrap tests between models. Cross-modal retrieval is image $\rightarrow$ note Recall@ K within each test fold (gallery = distinct notes in the fold, ≈ 120), averaged over folds.

Engineering. The training stack (train.py) uses PyTorch [35] DataLoader workers with pinned memory, persistent workers and prefetching; automatic mixed precision [36] (fp16 with loss scaling on GPU, bf16 autocast on CPU); DistributedDataParallel [37] launched with torchrun (NCCL on GPU, Gloo on CPU) with a DistributedSampler; and Weights & Biases [38] logging of every epoch, fold and seed (offline mode; synchronisable with wandb sync). All experiments in this paper were run on a 2-core CPU machine without a GPU; Section 6.7 reports the throughput measurements we could make there.

6 Results

6.1 Fusion helps, provided the text branch is competent

Table 1 and Figure 3 summarise the main comparison. Images alone reach 0.693 macro AUROC. Structured data alone reaches 0.657, the masked note alone 0.705, and note plus structured data 0.757. All three fusion paradigms that use every modality do better: early fusion 0.784, IRENE mid fusion ($k=2$) 0.785 and late fusion 0.794. A patient-level paired bootstrap (Table 2) finds IRENE clearly better than image-only (+0.117, 95% CI [+0.075, +0.155]). The gain over note + structured is smaller and borderline (+0.043, CI [-0.004, +0.091], $p=0.076$). The differences among the three fusion paradigms are within the bootstrap uncertainty (IRENE vs.

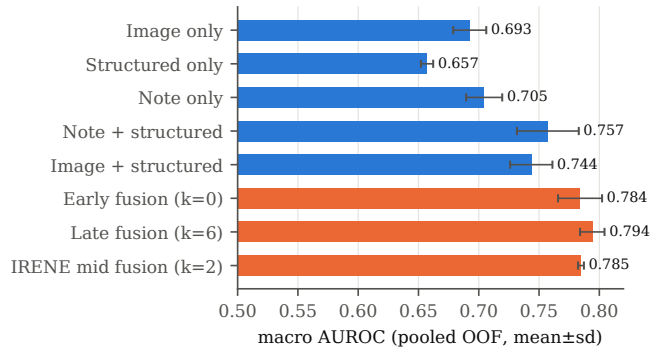


Figure 3: Macro AUROC by input and fusion paradigm. Blue: one or two modalities. Orange: all three modalities fused. Error bars: s.d. over 3 seeds.

early: $p=0.626$; vs. late: $p=0.768$). Early fusion is also the least stable across seeds (s.d. 0.018 against 0.002 for IRENE).

One engineering detail mattered more than any architectural choice. In a pilot run, the note branch was built only from static word vectors, truncated to 48 tokens and trained from scratch on about 500 notes. The note-only model then reached only about 0.58, and fused models about 0.71–0.74. A single *document token* fixes this: a TF-IDF $\rightarrow$ SVD embedding of the entire note, fitted inside each training fold, plays the role of IRENE’s BERT sentence features. With it and class-balanced BCE, the note-only model rises to 0.705 and the fused models to about 0.78–0.79. Table 3 isolates the document token (0.738 without it versus 0.785 with it).

6.2 Where to fuse

Figure 4 varies the number k of dual-stream blocks with IRENE’s bidirectional attention, keeping depth fixed at $L=6$. One or two blocks of cross-modal attention followed by joint self-attention ($k=1$: 0.780; $k=2$: 0.785) perform as well as early fusion. Accuracy then falls as fusion is delayed with cross-attention active ($k=3$: 0.770; $k=4$: 0.776). It collapses when all six blocks are bidirectional ($k=6$: 0.741). Yet the same depth *without* cross-attention, i.e. plain late fusion, is the best model (0.794). The pattern suggests that, on a few hundred training patients, the extra cross-attention parameters (Table 1) and the repeated mixing of a noisy note into the image stream cause overfitting rather than useful alignment. IRENE’s choice of shallow cross-modal attention followed by joint layers is a sensible point on this curve. The original paper, with about $10\times$ more patients, did not report this sweep.

6.3 How the streams attend to each other

Figure 5 fixes $k=2$ and changes only the combination rule. IRENE’s average of self- and cross-attention (0.785) matches the unidirectional variants (image $\rightarrow$ text 0.785, text $\rightarrow$ image 0.784) and the no-exchange control (0.779). It clearly beats ViLBERT-style co-attention, which drops self-attention in the dual-stream blocks (0.752). Flamingo-style tanh gates (0.777) are the least stable across seeds (s.d. 0.026). The practical

Table 1: Main comparison (5-fold patient-grouped CV, pooled out-of-fold predictions on all 782 images; mean $\pm$ s.d. over 3 seeds, or 2 seeds for the contrastive rows). The 95% CI is a patient-level bootstrap of the seed-averaged predictions. F1 and accuracy use the arg-max class. Parameters are trainable parameters, excluding the frozen DenseNet.

Model	macro AUROC	95% CI	macro AUPRC	macro F1	Accuracy	Params (M)
Image only	0.693 $\pm$ 0.014	[0.673, 0.758]	0.256 $\pm$ 0.007	0.242 $\pm$ 0.004	0.630 $\pm$ 0.005	1.00
Structured only	0.657 $\pm$ 0.005	[0.622, 0.711]	0.228 $\pm$ 0.005	0.177 $\pm$ 0.007	0.629 $\pm$ 0.004	1.00
Note only	0.705 $\pm$ 0.015	[0.678, 0.789]	0.383 $\pm$ 0.010	0.384 $\pm$ 0.038	0.649 $\pm$ 0.030	1.00
Note + structured	0.757 $\pm$ 0.026	[0.741, 0.843]	0.391 $\pm$ 0.035	0.351 $\pm$ 0.049	0.715 $\pm$ 0.014	1.00
Image + structured (k=2)	0.744 $\pm$ 0.018	[0.730, 0.810]	0.290 $\pm$ 0.009	0.245 $\pm$ 0.008	0.642 $\pm$ 0.015	1.46
Early fusion (k=0)	0.784 $\pm$ 0.018	[0.773, 0.871]	0.384 $\pm$ 0.018	0.368 $\pm$ 0.020	0.727 $\pm$ 0.012	1.00
Late fusion (k=6)	0.794$\pm$0.010	[0.785, 0.872]	0.399 $\pm$ 0.013	0.390 $\pm$ 0.033	0.719 $\pm$ 0.015	1.79
IRENE mid fusion (k=2)	0.785 $\pm$ 0.002	[0.795, 0.872]	0.369 $\pm$ 0.011	0.358 $\pm$ 0.024	0.710 $\pm$ 0.008	1.46
IRENE + CLIP ($\lambda=0.5$)	0.726 $\pm$ 0.003	[0.717, 0.797]	0.261 $\pm$ 0.001	0.228 $\pm$ 0.015	0.655 $\pm$ 0.004	1.53
IRENE + SigLIP ($\lambda=0.5$)	0.746 $\pm$ 0.018	[0.739, 0.813]	0.287 $\pm$ 0.020	0.274 $\pm$ 0.055	0.662 $\pm$ 0.012	1.53

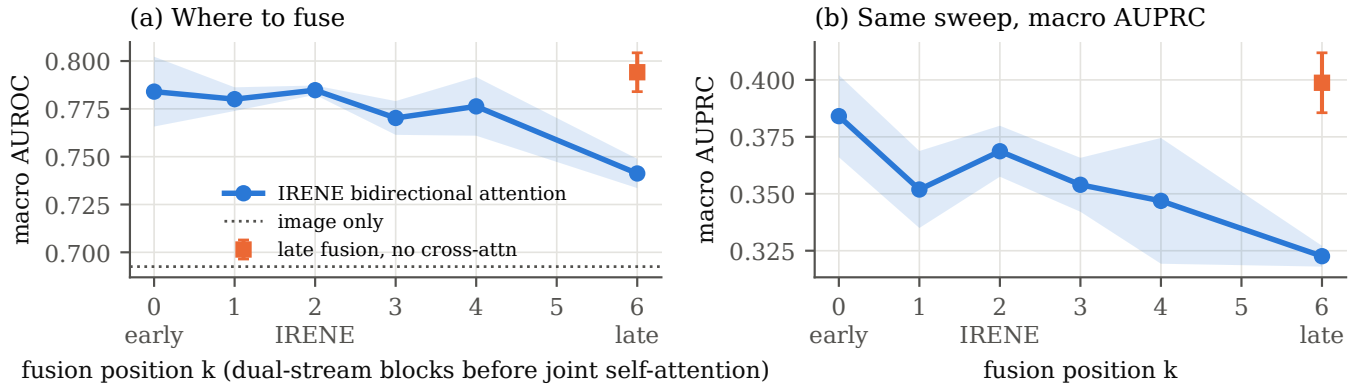


Figure 4: Fusion position. k dual-stream IRENE blocks are followed by $6 - k$ joint blocks; $k=0$ is early fusion. Orange square: $k=6$ without cross-attention (late fusion). Dotted line: image only. Shaded bands and error bars: ± 1 s.d. over seeds.

Table 2: Patient-level paired bootstrap (1,000 resamples) of the macro-AUROC difference between IRENE ($k=2$) and other models, using seed-averaged out-of-fold predictions.

IRENE ($k=2$)	Δ AUROC	95% CI	p
vs. Image only	+0.117	[+0.075, +0.155]	0.001
vs. Note + structured	+0.043	[-0.004, +0.091]	0.076
vs. Early fusion	+0.009	[-0.020, +0.047]	0.626
vs. Late fusion	+0.005	[-0.026, +0.038]	0.768
vs. No cross-attn ($k=2$)	+0.018	[-0.022, +0.058]	0.390
vs. IRENE + CLIP	+0.076	[+0.045, +0.107]	0.001

conclusion is that each stream must keep its intra-modal self-attention. Cross-attention in the first blocks is not harmful, but at this data scale the joint self-attention layers can recover most of what it provides. Figure 5b repeats the evaluation with the note removed. All variants fall to 0.58–0.61, which shows that the models rely heavily on the note whatever the attention structure.

6.4 Image–text contrastive learning

Figure 6 adds a CLIP or SigLIP loss on unimodal pooled embeddings. For IRENE ($k=2$), both objectives *lower* diagnostic AUROC, and more so as λ grows (CLIP: 0.739, 0.726, 0.717; SigLIP: 0.765, 0.746, 0.741). SigLIP degrades more gracefully than CLIP. Retrieval is above chance but modest ($R@5$

11.6% for CLIP at $\lambda=0.5$, against about 4% for chance). The picture changes for late fusion. The two encoders are then separate, so the model is a CLIP-style dual encoder with a classification head. Alignment leaves AUROC unchanged (0.793 against 0.794) and gives the best retrieval ($R@5$ 13.1%). In the early-fusion model the aligned embeddings sit at the token level, and alignment reaches 0.758.

Two properties of this dataset explain the weak benefit. First, there are only about 500 training pairs per fold, and many images of the same patient share one note, so the effective number of distinct pairs is small. Second, the notes describe history and radiographic findings in heterogeneous styles, with no one-to-one correspondence to the image. Contrastive pre-training works at the scale of hundreds of thousands to millions of report pairs [9, 10, 22]. As an auxiliary loss on hundreds of pairs, it competes with the supervised objective for the shared parameters in the dual-stream blocks, which is why it hurts mid fusion but not a dual encoder.

6.5 Per-class behaviour

Figures 7 and 8 break results down by class. Compared with the image-only model, fusion gains the most for COVID-19 (IRENE 0.87 vs. 0.75), bacterial pneumonia (0.76 vs. 0.64) and unspecified pneumonia (0.82 vs. 0.69). The note-based models are strong on exactly these classes (note + structured: 0.85, 0.82, 0.84). The image contributes more for fungal pneu-

Table 3: Ablations (pooled-OOF macro AUROC and AUPRC, mean $\pm$ s.d. over seeds). “No note” re-evaluates the same model with the note removed at test time. R@5 is image $\rightarrow$ note retrieval in % (chance $\approx$ 4%).

Variant	macro AUROC	macro AUPRC	AUROC, no note	R@5	seeds
<i>Fusion position</i>					
k=0 (early)	0.784 $\pm$ 0.018	0.384 $\pm$ 0.018	0.621 $\pm$ 0.013	–	3
k=1	0.780 $\pm$ 0.006	0.352 $\pm$ 0.017	0.566 $\pm$ 0.000	–	2
k=2 (IRENE)	0.785 $\pm$ 0.002	0.369 $\pm$ 0.011	0.588 $\pm$ 0.003	–	3
k=3	0.770 $\pm$ 0.009	0.354 $\pm$ 0.012	0.571 $\pm$ 0.006	–	2
k=4	0.776 $\pm$ 0.015	0.347 $\pm$ 0.028	0.583 $\pm$ 0.008	–	2
k=6, bidirectional	0.741 $\pm$ 0.008	0.323 $\pm$ 0.005	0.586 $\pm$ 0.022	–	2
k=6, no cross-attn (late)	0.794 $\pm$ 0.010	0.399 $\pm$ 0.013	0.655 $\pm$ 0.019	–	3
<i>Attention structure (k=2)</i>					
self only (no exchange)	0.779 $\pm$ 0.002	0.357 $\pm$ 0.028	0.611 $\pm$ 0.001	–	2
image $\rightarrow$ text	0.785 $\pm$ 0.005	0.383 $\pm$ 0.004	0.582 $\pm$ 0.019	–	2
text $\rightarrow$ image	0.784 $\pm$ 0.008	0.361 $\pm$ 0.008	0.594 $\pm$ 0.006	–	2
co-attention (cross only)	0.752 $\pm$ 0.005	0.324 $\pm$ 0.006	0.591 $\pm$ 0.006	–	2
gated cross-attn	0.777 $\pm$ 0.026	0.371 $\pm$ 0.014	0.587 $\pm$ 0.012	–	2
IRENE (self+cross)/2	0.785 $\pm$ 0.002	0.369 $\pm$ 0.011	0.588 $\pm$ 0.003	–	3
<i>Alignment objective (k=2)</i>					
none	0.785 $\pm$ 0.002	0.369 $\pm$ 0.011	0.588 $\pm$ 0.003	–	3
CLIP λ =0.1	0.739 $\pm$ 0.004	0.297 $\pm$ 0.033	0.618 $\pm$ 0.025	10.9	2
CLIP λ =0.5	0.726 $\pm$ 0.003	0.261 $\pm$ 0.001	0.617 $\pm$ 0.011	11.6	2
CLIP λ =1.0	0.717 $\pm$ 0.000	0.256 $\pm$ 0.004	0.620 $\pm$ 0.006	10.7	2
SigLIP λ =0.1	0.765 $\pm$ 0.014	0.336 $\pm$ 0.022	0.584 $\pm$ 0.016	9.1	2
SigLIP λ =0.5	0.746 $\pm$ 0.018	0.287 $\pm$ 0.020	0.616 $\pm$ 0.001	10.5	2
SigLIP λ =1.0	0.741 $\pm$ 0.005	0.281 $\pm$ 0.017	0.624 $\pm$ 0.018	9.7	2
CLIP λ =0.5, early (k=0)	0.758 $\pm$ 0.006	0.322 $\pm$ 0.002	0.648 $\pm$ 0.006	12.1	2
CLIP λ =0.5, late (k=6)	0.793 $\pm$ 0.004	0.322 $\pm$ 0.010	0.699 $\pm$ 0.005	13.1	2
<i>Note input and training (k=2)</i>					
default (word tokens + doc token, masked)	0.785 $\pm$ 0.002	0.369 $\pm$ 0.011	0.588 $\pm$ 0.003	–	3
without document token	0.738 $\pm$ 0.013	0.299 $\pm$ 0.016	0.599 $\pm$ 0.010	–	2
raw notes (diagnosis terms visible)	0.794 $\pm$ 0.008	0.387 $\pm$ 0.020	0.587 $\pm$ 0.008	–	2
modality dropout p =0.3	0.772 $\pm$ 0.023	0.339 $\pm$ 0.016	0.736 $\pm$ 0.011	–	2

monia (image-only 0.78 vs. note + structured 0.74), where fusion combines both (0.86). Tuberculosis and no finding (18 images each) remain the hardest and noisiest classes for every model.

6.6 Label leakage and dataset confounding

Diagnosis terms in notes. 52% of the notes name the diagnosis or the pathogen. Leaving these terms unmasked raises note + structured from 0.757 to 0.783 and IRENE from 0.785 to 0.794 (Figure 9a). The gain is smaller than one might expect, because the remaining text is itself highly predictive. A TF-IDF logistic regression on the *masked* notes reaches 0.831, almost as high as on raw notes (0.836), and above every transformer in this study (Table 4). Masking is still necessary for a fair evaluation, but it is far from sufficient.

Source confounding. Public COVID-19 collections gather cases from sources that differ systematically in their label mix. Some are almost entirely COVID-19 (the SIRM archive, GitHub-hosted hospital releases, NEJM case reports), while others are mostly non-COVID (Eurorad). A logistic regression that sees *only* the host of the source URL reaches 0.740

macro AUROC, on par with the fusion transformers. The notes and the view (0.601) carry this signal implicitly through writing style, translation artefacts and acquisition protocol. As a de-confounded check, we recompute every metric on the 395 images from Radiopaedia and Eurorad, the two repositories that host both COVID-19 and non-COVID cases. There the source-only probe falls to chance (0.487). The fused models lose about 6 points and the image-only model about 3, but the ranking is preserved (Table 4). Late fusion (0.729), early fusion (0.724) and IRENE (0.723) remain above note + structured (0.686) and image-only (0.662). The TF-IDF note model is still the strongest (0.778). This agrees with the radiograph shortcuts reported by DeGrave et al. [12] and argues for reporting a de-confounded subset alongside headline numbers on public COVID-19 data.

Missing notes. Deployed systems will not always have a note. When it is removed at test time, IRENE falls from 0.785 to 0.588, well below the image-only model (0.693). Early and late fusion degrade less (0.621 and 0.655; Figure 9b). The fused models have learned to rely on the note and do not fall back on the image features. This matches the fragility reported by Ma et al. [19]. Training with modality dropout

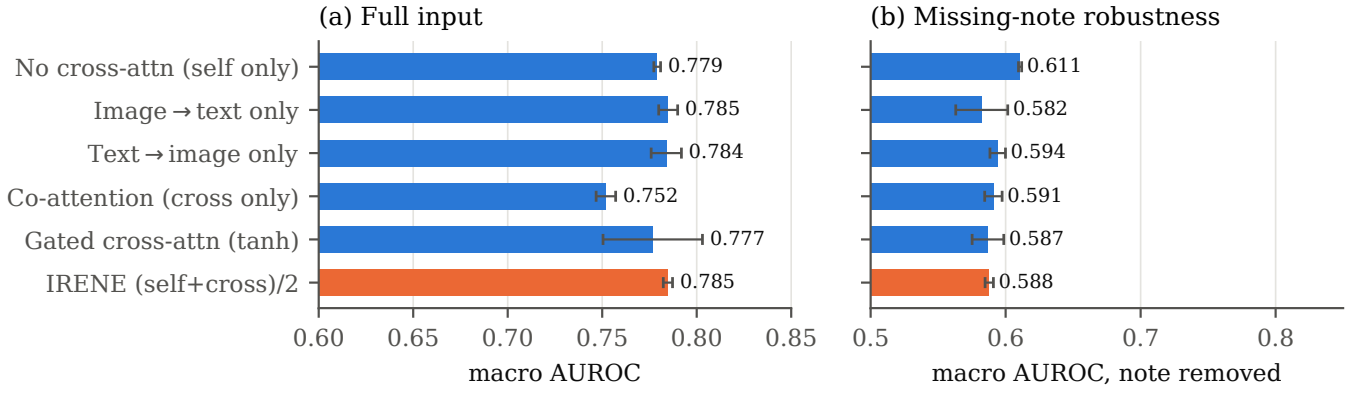


Figure 5: Attention structure inside the $k=2$ dual-stream blocks. (a) Full input. (b) The same trained models evaluated with the clinical note removed.

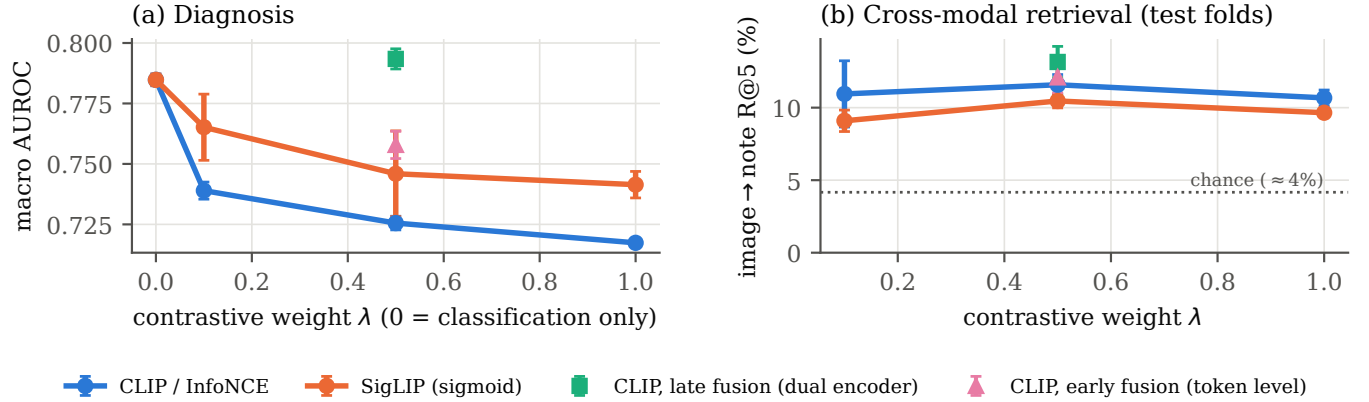


Figure 6: Contrastive alignment. (a) Diagnostic macro AUROC as a function of the alignment weight λ ($\lambda=0$ is IRENE without alignment). (b) Image→note Recall@5 on the test folds.

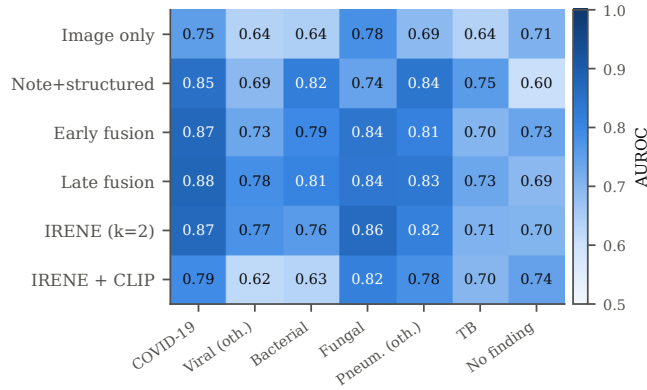


Figure 7: Per-class AUROC (pooled OOF, mean over seeds).

($p=0.3$) raises the no-note AUROC to 0.736, close to the image + structured model (0.744), at a small cost with full input (0.772). For a clinical deployment where the note is not always available, this trade-off is clearly worthwhile.

6.7 Throughput of the training stack

Figure 10 reports the engineering measurements we could make without a GPU. Many source radiographs are large JPEG/PNG files, and decoding and resizing them for every sample limits the input pipeline to 46 samples/s in the main process. Two worker processes help ($\times 1.9$, 85 samples/s).

Pre-resizing once into a memory-mapped uint8 array takes 19 s and raises throughput to 7324 samples/s ($\times 161$). With so little per-sample work on 2 cores, worker processes then *reduce* throughput because of inter-process transfer. The right `num_workers` depends on the per-sample cost, and should be measured rather than set by habit. Persistent workers avoid re-forking at every epoch and help consistently when workers are used. For the model, `bf16` autocast raises end-to-end training throughput from 11.7 to 15.5 samples/s on this CPU. DDP with two single-threaded Gloo ranks reaches 14.7 samples/s, against 11.7 for one process with two threads. The same `train.py` runs unchanged with NCCL and `fp16` loss scaling on GPUs, but we could not measure GPU utilisation here and make no claim about it.

7 Discussion

What matters for IRENE-style fusion at small scale. Three findings are consistent across metrics and seeds. (1) *Fusion helps, and the text representation decides by how much.* Adding the note and structured data to the image raises macro AUROC by about 0.09–0.10. However, a weak note encoder (static word vectors alone) wiped out most of this gain. A document-level token recovered it. (2) *Keep intra-modal self-attention, and limit cross-attention to early layers.* IRENE’s

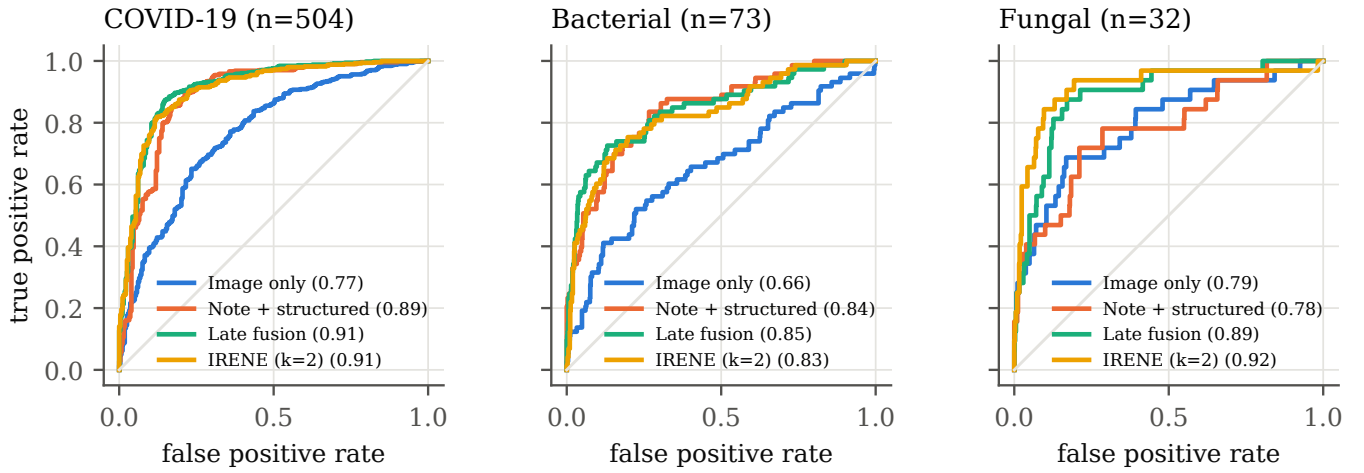


Figure 8: Pooled out-of-fold ROC curves (seed-averaged predictions) for three classes; AUROC in the legend.

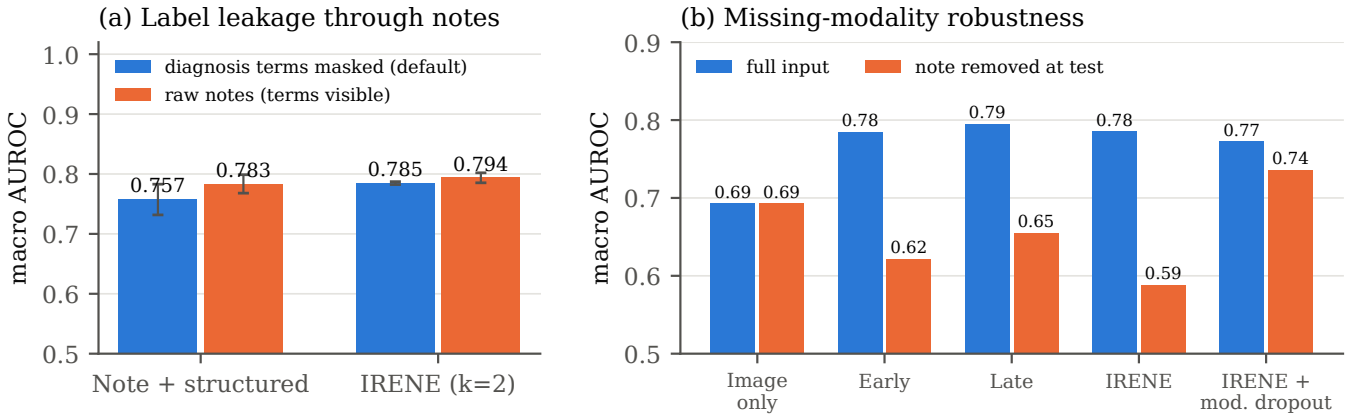


Figure 9: (a) Effect of masking diagnosis and pathogen names in the notes (2 seeds for the raw-note runs). (b) Macro AUROC with the full input and with the note removed at test time. Modality dropout ($p=0.3$) during training closes most of the gap.

(self+cross)/2 blocks followed by joint self-attention are on par with early and late fusion. Cross-attention without self-attention, or cross-attention at every layer, is clearly worse. On a few hundred patients, extra cross-modal capacity causes overfitting. (3) *Contrastive alignment is not free*. As an auxiliary loss on hundreds of pairs, CLIP and SigLIP hurt a mid-fusion model and are neutral for a dual encoder. The benefit reported for medical vision–language models comes from large-scale pre-training, which our setting lacks. A natural next step is to initialise the two streams from a publicly released radiograph–report encoder (e.g. BiomedCLIP or CXR-CLIP) instead of learning the alignment on the target data.

What the headline numbers hide. The dataset is public, multimodal and, with IRENE’s input types, a natural testbed. It is also strongly confounded by source. A source-only probe matches the fusion transformers, and a linear bag-of-words model on the masked notes beats all of them. The de-confounded subset preserves the ranking of the fusion strategies, which supports the architectural conclusions. The absolute numbers, however, should not be read as clinical diagnostic accuracy.

8 Limitations

- **Scale.** 782 images from 424 patients, and three or fewer positives per test fold for some classes. We pool out-of-fold predictions and report bootstrap intervals. Many ablation differences are nevertheless within noise, and the ablations use 2 seeds.
- **Compute.** No GPU was available. The model is a narrow IRENE ($d=128$, 6 layers, 16 image tokens from a frozen CNN) rather than the original ViT-B/16 initialised from ImageNet weights, and the text encoder is not a clinical BERT. The recipe was revised once, after a full cross-validation pilot showed that the note branch was the bottleneck (it added the document token and class-balanced loss). This single round of development used out-of-fold results, so a fresh external test set would be needed for an unbiased estimate.
- **Data.** The labels come from publication metadata, the notes are written by case authors rather than being contemporaneous EHR text, and the source confounding described in Section 6.6 limits clinical interpretation. The engineering code supports large sharded image–text corpora and multi-GPU training, but this study did not exercise that scale.

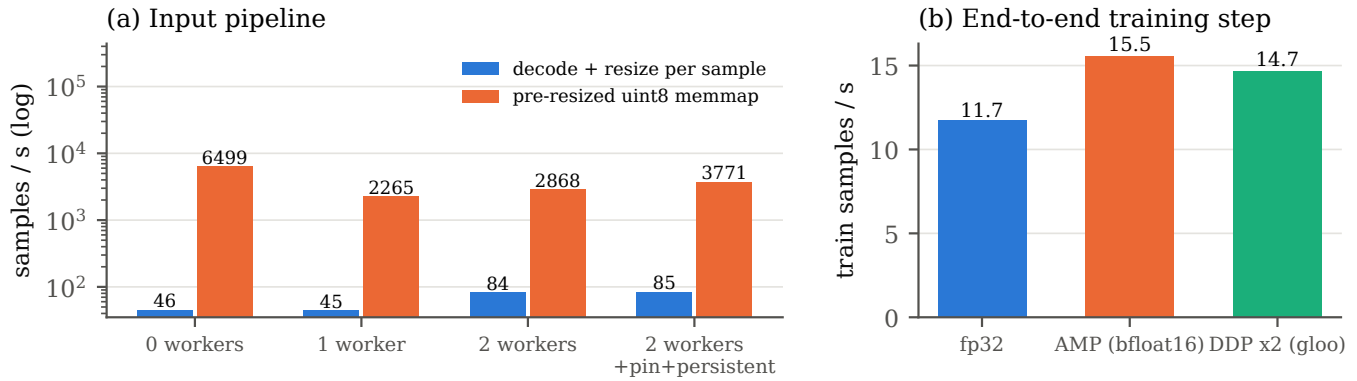


Figure 10: Throughput on the 2-core CPU machine used for all experiments. (a) Input pipeline: decoding and resizing the original radiographs per sample, versus a pre-resized uint8 memory-mapped cache, for several `DataLoader` settings. (b) One end-to-end training step of pixel-input IRENE-MM (batch 16), in fp32, with bf16 autocast, and with DDP over 2 Gloo processes (1 thread each).

Table 4: Shortcut probes. Macro AUROC on all 782 images and on the 395 images from the two case repositories that contain *both* COVID-19 and non-COVID cases (Radiopaedia, Eurorad), where the source cannot identify the label. Logistic regressions use the same folds.

Model	all images	mixed-source subset
<i>Logistic regression probes</i>		
Source host only (one-hot)	0.740	0.487
View only	0.601	0.570
All structured fields	0.737	0.670
TF-IDF note, [DX]-masked	0.831	0.778
TF-IDF note, raw	0.836	0.780
<i>IRENE-MM variants (mean over seeds)</i>		
Image only	0.693	0.662
Note + structured	0.757	0.686
Early fusion	0.784	0.724
Late fusion	0.794	0.729
IRENE (k=2)	0.785	0.723
IRENE + CLIP	0.726	0.683

9 Conclusion

We provide an open, end-to-end re-implementation of the IRENE multimodal transformer on public data. It includes a documented image–text–structured pipeline, a modern PyTorch training stack, a single model that covers early, mid and late fusion, six attention structures and two contrastive objectives, and a controlled study with patient-grouped cross-validation. The results support IRENE’s core design: shallow bidirectional attention that retains self-attention, followed by joint layers. They also show that a competent text representation, robustness to missing notes and awareness of dataset confounding matter more than the particular choice of fusion operator.

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A Reproducibility details

Code. Everything is in the repository: `data/prepare.py` (cohort, labels, text cleaning, masking, structured features, folds), `data/features.py` (frozen CXR DenseNet tokens with cached augmentations; word-vector note tokens), `models/irene_mm.py` (IRENE-MM), `train.py` (CV training with AMP/DDP/W&B), `scripts/run_experiments.py` (the full grid), `scripts/classical_baselines.py`, `scripts/benchmark_throughput.py` and `scripts/make_figures.py` (all figures, tables and every number quoted in the text, emitted as LaTeX macros). The original IRENE files in `models/` and `irene.py` are kept, with the deprecated apex AMP replaced by `torch.autocast`.

Pretrained components. Image: `torchxrayvision` DenseNet-121 `densenet121-res224-all`, trained on NIH ChestX-ray14, CheXpert, MIMIC-CXR, PadChest, RSNA Pneumonia (Kaggle), Google and OpenI radiographs [32]; none of these is the evaluation dataset, but overlap of individual source images cannot be excluded. Text: `spaCy` `en_core_web_md` 300-d vectors (20k keys). No component was pretrained on the evaluation notes.

Hyper-parameters. $L=6$, $d=128$, 4 heads, MLP 256, dropout and attention dropout 0.1, image grid 4×4 (16 tokens), at most 48 word tokens plus one 128-d TF-IDF/SVD document token fitted inside each training fold, 11 structured tokens plus age and sex tokens, AdamW ($\text{lr } 3\times 10^{-4}$, $\text{wd } 0.05$), one-cycle schedule with 10% warm-up, batch 32, ≤ 25 epochs, early stopping on validation BCE with patience 5, positive-class weights $\sqrt{n_{\text{neg}}/n_{\text{pos}}}$ in the BCE, gradient clipping 1.0, CLIP temperature initialised at 0.07, SigLIP $t=10$, $b=-10$ [6]. No per-configuration tuning was done. However, the recipe was revised once: a first full cross-validation pilot of the eight

main configurations (kept in `results/runs_v0_pilot/`) showed that the note branch was the bottleneck, and we then added the document token and the class-balanced loss and reran everything. Because that decision was informed by out-of-fold results, the reported estimates are slightly optimistic in the usual “one round of model development” sense (Section 8).

Compute. All 29 configurations (330 fold-level trainings) ran in 10.1 CPU-hours on a 2-core virtual machine without a GPU.